VolDrive

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Context first. Pressure second. Execution last.

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1. Overview

prevailing trend, VolDrive prints an arrow. When the market is just drifting, chopping, or printing small indecisive bars, it stays quiet.

The thinking behind it is simple: a move worth joining usually has two things at once. There is **effort** (participation, seen in volume) and there is **result** (price genuinely moving, not just wicking back and forth). A lot of volume that goes nowhere is absorption, not drive. A clean move on no volume is thin and unreliable. VolDrive looks for the combination — effort that produces movement — and only in the direction the market is already trending.

To keep its arrows meaningful, VolDrive runs each candidate through a short stack of **gates**. Each gate asks one question — is the market trending cleanly, which way, is there real push, is it consistent, is this bar decisive, are we stuck in a range — and only when the gates agree does an arrow appear. The more gates you enable, the fewer but higher-quality the signals. VolDrive is a context tool: it highlights where continuation is likely, but the entry, exit and risk decisions remain yours.

For speed, VolDrive can read a fast **seconds series** for the responsive parts of its work while confirming and drawing on your primary chart bar. That same arrow is what the BloodHound and BlackBird outputs expose for automation.

2. What you see on the chart

The core visual is an **arrow** at a bar where all the active gates passed — pointing up for a long-side continuation, down for a short-side one. Arrow size, offset, colors and outline are all configurable, and the arrow's opacity reflects the strength behind it, so stronger signals stand out and marginal ones are fainter.

Optionally, VolDrive can **gray out chop bars** — a purely visual aid that shades the sideways congestion it detects, so you can see at a glance where it is suppressing signals. And when **tooltips** are on, hovering over any arrow shows its detail: the energy reading, the sequence state, the volume context, and the **auction phase** of the move (see Section 6).

That is the whole footprint — arrows where continuation is likely, optional gray congestion shading, and rich hover detail when you want it.

3. The signal pipeline

VolDrive evaluates each bar through a pipeline of gates. If any active gate says no, no arrow is printed. In order:

1. **Conditions** — is the market trending cleanly enough to trade? (efficiency, trend fit, trend strength)
2. **Trend** — which direction is the move, and has that direction been stable?
3. **Energy** — is there real push: volume effort producing a price result?
4. **Sequencing** — has that push shown up consistently, not just once?
5. **Doji filter** — is this bar decisive, or a small indecisive candle?
6. **ChopState filter** — are we stuck inside a sideways box?
7. **Session** — are we inside an allowed trading window?

A key design point: the conditions gate does not freeze the rest of the engine. VolDrive keeps its trend, energy and sequence state warm at all times and uses the conditions check only as a final "may I print an arrow now?" gate. That avoids the trap where turning the conditions filter on makes everything else go cold. When **Regime On Primary** is enabled, that final check is read from your calmer primary timeframe even while the faster work runs on the seconds series.

4. The gates, one by one

4.1 Conditions — the market's weather

Before looking for a push, VolDrive checks whether the market is trending cleanly enough to bother. It can do this two ways, chosen by **Regime Mode**: either a combined check of efficiency, trend fit and trend strength (all three must pass), or a simpler trend-strength-only check.

In plain terms, the three readings ask: is the move reasonably straight rather than wandering (efficiency), does a straight trend line describe it well (trend fit), and is there enough directional strength behind it (trend strength). When the conditions filter is on, arrows only appear while these pass. When it is off, conditions are ignored entirely. The

check is a gate on *printing* — VolDrive keeps tracking everything else underneath regardless.

4.2 Trend — the direction

The trend gate decides which way is "up" for signals and tries not to change its mind too easily. Two engines are available via **Trend Engine**: an EMA-slope reading (direction follows the slope of a moving average) or a Donchian breakout reading (direction flips when price clears the high or low of a lookback window by a buffer).

Two settings keep the direction stable. **Confirm Seconds Bars** requires several seconds-bars to agree before accepting a flip, and **Min Bars After Flip** imposes a cooldown of primary bars after a direction change before a new signal is allowed. Together they stop the rapid flip-flopping that produces noise.

4.3 Energy — the strength of the push

Energy is the heart of VolDrive. It combines **effort** and **result**. Effort grows when volume is high relative to its own baseline and when that volume is packed into a small price range. Result grows when the candle body is substantial relative to its full range — a real body, not all wicks — and when there is clean momentum in the trend direction. Energy is effort and result together: a hard push that actually moves price.

Two thresholds exist because the seconds series and the primary fallback have different data shapes: **Energy Min (Seconds Slot)** applies when the fast series is active, and **Energy Min (Backfill Slot)** applies when VolDrive falls back to the primary bar. **Use Energy Filter** decides whether energy is a hard gate. When on, the signal bar itself must clear the threshold, which tends to produce later, more climactic signals. When off, energy is still calculated, shown in tooltips and fed to sequencing, but it does not block arrows — which tends to produce earlier, continuation-style signals.

4.4 Sequencing — consistency over a one-off

Sequencing asks whether strong pushes have shown up *repeatedly*, not just on a single bar. It looks back over a window of **Sequence Window (N)** bars and counts how many were energy hits; the gate passes when at least **Required Hits (M)** of them qualify. This filters

out lone random spikes and favors markets where energy keeps appearing. **Min Bars Between Signals** then spaces arrows out on the primary chart so they do not cluster in one spot.

Energy and sequencing work well together. With both on, you are asking for a strong push now *and* a recent history of them — very selective. With energy off and sequencing on, you are allowing continuation signals when enough strong pushes happened recently even if this exact bar is not the spike — often the sweet spot for earlier entries.

4.5 The doji filter — skip indecisive bars

A doji is a bar with a small body and large wicks — the market saying "maybe." The doji filter blocks arrows on bars whose body is too small relative to their full range, set by **Doji Max Body (% of TR)**. The check runs twice — once on the active series and again before the final commit on the primary bar — so a bar that looks fine on the fast series but maps to a noisy primary bar is still rejected.

4.6 The ChopState filter — stay out of boxes

Sometimes the market is a ball bouncing in a small room: lots of motion, no progress. The ChopState filter detects that sideways congestion and suppresses signals while price is inside a box, keeping arrows out of the whippiest areas. It judges congestion from progress and bar overlap, and it has its own breakout rules that define when price has genuinely escaped the box and for how long to treat that as a breakout. If you are unsure, simply leave the filter on or off and tune the rest first; the breakout parameters are there for fine control when you want it. There is also an optional **VolSpike suppression** that hides an arrow on any bar where Flow.42 VolSpike has flagged a volume spike, when that tool is on the same chart.

5. The Signal Sensitivity preset

Most users do not need to touch the advanced groups. **Signal Sensitivity** is a single dial that sets the detection thresholds for you:

- **Conservative** — fewer, cleaner signals. Demands stronger conditions, more energy, and more consistency.
- **Balanced** — the shipped defaults, a middle ground.
- **Aggressive** — more, earlier signals, at the cost of some quality.
- **Custom** — leaves every value exactly as you set it in the advanced groups. This is the default, so existing setups behave unchanged.

Start with a preset that matches how often you want to see arrows, then, if you like, switch to Custom and fine-tune individual gates. Each advanced group also has a **Guided Tuning** panel that lets you adjust that gate in plain language with explained sliders instead of raw numbers.

6. Auction phase and the Auction Engine

Every VolDrive arrow carries an **auction phase** in its tooltip, classifying the move's maturity as **Ignition** (a push just getting going), **Drive** (a push in full force), or **Exhaustion** (a push running out of fuel). This helps you tell an early continuation from a late, stretched one at a glance.

Separately, if Flow.42 AuctionEngine is on the same chart, VolDrive can let the broader auction context shade its arrows. **Auction Modulation** keeps arrows aligned with the auction thesis at full opacity while dimming counter-trend ones, so the highest-confluence signals stand out; set it to Off for no dimming (the engine still appears in the tooltip). **Auction Context** chooses whether that dimming uses the engine's current read (Live) or the read as it was at each arrow's own bar (Historical, useful for studying past behavior, even on a closed market). With no AuctionEngine on the chart, this group has no effect.

7. Sessions, sounds and outputs

Sessions. VolDrive can limit arrows to time windows — a Pre-Market window and an RTH window, each switchable with its own start and end time. With **Ignore Session Filter** on, the

windows are bypassed and signals can occur at any time. Times accept both 24-hour (`15:00`) and 12-hour (`3:00 PM`) formats in your local time zone.

Sound alerts. With **Sound Alerts** on, VolDrive plays a chosen WAV file for long signals and another for short signals, drawn from NinjaTrader's sounds folder. An optional **Rearm** prevents repeated alerts within a set number of seconds.

Automation outputs. VolDrive exposes three outputs for BloodHound and BlackBird: `VolDriveLong` pulses on a long signal, `VolDriveShort` pulses on a short signal, and `VolDriveSignal` carries +1, −1 or 0. These are calculated on bar close for stable, non-repainting behavior on reload.

8. Full parameter reference

The settings appear in the NinjaTrader properties panel, grouped as you see them there. "Default" is the value the indicator ships with. The detailed numeric controls live in the Advanced groups; the filter on/off switches live together in 01 Filters.

00 Data

Setting	Default	What it does
Signal Sensitivity (preset)	Custom	One global dial that tunes the detection thresholds. Custom keeps your manual values; Conservative, Balanced and Aggressive apply progressively looser threshold sets. See Section 5.
Seconds Bars Period	10	The period of the fast secondary seconds series used for the responsive parts of the signal. If no seconds history is available, VolDrive falls back to the primary series.

01 Filters

Setting	Default	What it does
Use Regime Filter	Off	When on, arrows only print while the market-conditions check passes. When off, conditions are ignored entirely.
Regime On Primary	On	Reads the conditions check from your primary chart timeframe (calmer, more stable) even when the seconds series drives energy, trend and sequencing.
Use Energy Filter	Off	When on, the signal bar itself must clear the energy threshold (a hard gate). When off, energy is still calculated and feeds tooltips and sequencing but does not block arrows.
Use Sequence Filter	On	When on, signals must satisfy the M-of-N sequence rule. When off, sequencing is ignored.
Use Doji Filter	On	Discards bars whose body is too small relative to their range. Checked on the active series and again before commit on the primary bar.
Doji Max Body (% of TR)	0.26	The body-to-range threshold for the doji filter. Lower is stricter (fewer small bars pass). Typical range 0.20–0.30.
Use ChopState Filter	On	Suppresses signals inside confirmed sideways congestion (chop boxes).
ChopState Candles Gray	On	A visual aid only — shades detected chop bars gray. No effect on logic.
Suppress Arrow on VolSpike	Off	When on, hides an arrow on any bar where Flow.42 VolSpike flagged a volume spike. Requires VolSpike on the same chart.

02 CHOP Breakout

These control how the chop detector decides a market is in a box and what counts as breaking out of it. If unsure, leave them at default and use the ChopState filter on/off first.

Setting	Default	What it does
ER Period	50	Efficiency lookback inside the chop model. Higher means slower, fewer state changes.

Setting	Default	What it does
ER Low	0.35	Below this efficiency, the model leans toward chop.
ER High	0.60	Above this efficiency, the model leans toward trend.
Overlap Lookback K	8	How many bars are checked for tight overlap (sideways compression).
Overlap Threshold	0.71	How much overlap counts as tight. Higher is stricter, producing fewer boxes.
Small ATR Factor	0.80	Bars larger than this multiple of ATR are treated as too big for the overlap test, so one large bar cannot distort it.
Progress Frac Max	0.25	The most close-to-close progress, as a fraction of ATR, still allowed while counting as chop.
Overlap ER Ceiling	0.45	Overlap only counts while efficiency is below this, keeping the logic focused on slow zones.
ATR Period	14	The volatility lookback used by the chop model.
Consecutive Bars M	3	Bars required to confirm entering or leaving the chop state (anti-flicker).
Enter Score	1	The score that starts counting toward chop.
Exit Score	0	The score that starts counting toward trend.
Chop Trimmer	—	Opens the Chop Trimmer editor to configure the breakout parameters with presets.
Breakout Lookback	10	Lookback that defines the box edges for the breakout check.
Big Bar Factor	0.75	A bar must be at least this multiple of ATR to qualify as a strong breakout bar.
Breakout Hold Bars	3	How long the breakout (trend-allowed) state is held after a breakout.
Box Edge Buffer (Ticks)	6	Extra ticks around the box edges before declaring a breakout, reducing false breaks.

Setting	Default	What it does
Big Body Factor Patch	1.00	An extra body-size rule used by the breakout confirmation.
Disable Overlap Bars	3	Bars to temporarily suspend overlap scoring after a breakout, preventing immediate re-boxing.
Hold Gamma	1.80	A dynamic hold multiplier — bigger breakout bars can hold the state longer.
Box Stickiness ATR	0.30	How far, in ATR, the box edges may stretch, for stability.

03 Session Settings

Setting	Default	What it does
Ignore Session Filter	Off	When on, all session windows are bypassed and signals can occur at any time.
Use Pre-Market	On	Enables the Pre-Market session window.
Pre Market Start (HH:mm)	09:00	Pre-Market start time, in your local time. Accepts 24-hour and 12-hour formats.
Pre Market End (HH:mm)	13:00	Pre-Market end time, in your local time.
Use RTH	On	Enables the Regular Trading Hours window.
RTH Start (HH:mm)	15:00	RTH start time, in your local time.
RTH End (HH:mm)	22:00	RTH end time, in your local time.

04 Visuals

Setting	Default	What it does
Arrow Size (px)	10	The size of the signal arrow in pixels.

Setting	Default	What it does
Arrow Offset (px)	12	The distance the arrow sits from the bar.
Min Arrow Opacity	0.25	The opacity of the weakest signals. Arrow opacity scales with signal strength.
Max Arrow Opacity	1.00	The opacity of the strongest signals.
Arrow Up Color	LimeGreen	The color of long (up) arrows.
Arrow Down Color	OrangeRed	The color of short (down) arrows.
Arrow Outline Color	White	The outline color of the arrows.

05 Sound Alerts

Setting	Default	What it does
Sound Alerts	Off	Enables sound alerts for VolDrive signals.
VolDrive LONG sound file	Flow42alert01.wav	The WAV file in NinjaTrader's sounds folder played for long signals.
VolDrive SHORT sound file	Flow42alert02.wav	The WAV file played for short signals.
Rearm	Off	Uses NinjaTrader's rearm timing to avoid repeated alerts within a set interval.
Rearm Time (s)	30	The rearm interval in seconds, used only when Rearm is on.

06 Tooltips

Setting	Default	What it does
Enable Tooltips	Off	Shows a hover tooltip with energy, sequence, volume and auction-phase detail for each arrow.
Marker Hitbox Size (px)	14	The size of the invisible hover area that triggers the tooltip.

07 Auction Engine

These settings only do something when Flow.42 AuctionEngine is on the same chart.

Setting	Default	What it does
Auction Modulation	Subtle	Dims arrows that run counter to the auction thesis while keeping aligned arrows full opacity. Off means no dimming (the engine still appears in the tooltip). Options: Off, Subtle, Strong.
Auction Context	Live	Whether the dimming uses the engine's current read (Live) or the read as it was at each arrow's own bar (Historical).

08 Help & Hotkey

Setting	Default	What it does
Show Help Panel	—	Shows or hides the on-chart help panel.
Enable Help Hotkey	—	Lets you toggle the help panel with a keyboard shortcut.
Assigned Help Hotkey	CTRL + H	The shortcut that toggles the help panel. Read-only.
Help Panel Width (px)	—	The width of the help panel in pixels.
Help Panel Height (px)	—	The height of the help panel in pixels.
Help Panel Opacity (%)	—	The opacity of the help panel, from 0 to 100.

09–12 Advanced (Conditions, Energy, Sequencing, Trend)

The advanced groups hold the detailed numeric controls behind the filters. Each also offers a **Guided Tuning** panel that lets you adjust the gate in plain language instead of raw numbers.

Setting	Default	What it does
Regime Mode (Trend Filter)	ER+R²+ADX	How market conditions are judged: the combined efficiency-plus-fit-plus-strength check, or strength only.
Efficiency Lookback (ER Length)	20	Bars used to measure how straight the move is. Higher is slower and steadier.
Efficiency Threshold (ER Min)	0.35	The minimum straightness for conditions to pass. Higher lets only cleaner trends through.
Trend Fit Lookback (LR Length)	50	Lookback for the trend-fit measure. Higher is smoother.
Trend Fit Min (R² Min)	0.30	The minimum trend fit for conditions to stay on. Higher accepts only well-behaved trends.
ADX Period	14	The trend-strength lookback. Higher is slower and steadier.
ADX Min Level	18	The minimum trend strength for conditions to pass. Higher requires stronger trends.
Trend Engine	EMA Slope	Which engine sets direction: EMA slope, or Donchian breakout.
Donchian Lookback	20	The channel length for the Donchian breakout engine.
Breakout Buffer (ticks)	4	Extra ticks beyond the channel to trigger a breakout. Higher gives fewer, cleaner breaks.
EMA Slope Lookback	13	The EMA length whose slope sets direction in EMA-slope mode.
Confirm Seconds Bars	2	Consecutive seconds-bars that must agree before a trend flip is accepted.

Setting	Default	What it does
Min Bars After Flip	2	Primary bars to wait after a direction flip before a new signal is allowed.
Volume EMA Length	50	The baseline-volume length inside the energy calculation. Higher is smoother.
VPR Threshold (Percentile)	0.60	The percentile at which volume-per-range counts as high. Higher demands stronger volume spikes.
Weight: Z-Score Volume (wZV)	0.6	How much raw volume surges contribute to effort.
Weight: Volume per Range (wVPR)	0.4	How much volume packed into a small range contributes to effort.
Result Blend (Body vs Momentum)	0.5	The mix between candle-body result and momentum result (0 = all momentum, 1 = all body).
Energy Min (Seconds Slot)	0.35	The minimum energy for a hit on the fast seconds series.
Energy Min (Backfill Slot)	0.20	The minimum energy when VolDrive falls back to the primary bar. Usually lower.
Volume Window (bars)	240	The window used to measure the volume baseline and percentiles. Higher is more stable.
Sequence Window (N)	3	The number of recent bars over which energy hits are counted.
Required Hits (M)	1	The minimum energy hits inside the window for sequencing to pass. Higher is more selective.
Min Bars Between Signals	3	The minimum spacing in primary bars between two signals.

9. Suggested starting settings

The fastest path is the **Signal Sensitivity** preset (Section 5): pick Conservative, Balanced or Aggressive to match how many arrows you want, and adjust from there. Below are rough starting points by timeframe, expressed as targets rather than rigid values — validate on your own data before committing capital. Session times are examples in Central European Time; adjust to your zone.

Faster charts (1–2 minute). Expect a higher signal count. Keep the seconds series short, conditions thresholds moderate, and energy permissive (energy filter off) so continuation signals appear earlier. Aim for a handful of signals per hour and tighten with Required Hits or Min Bars Between Signals if it feels noisy.

Mid charts (5 minute). A longer seconds series and slightly stricter conditions suit the slower pace. Expect one to three signals per hour. This is a comfortable default zone for most index futures.

Slower charts (10–15 minute). Lengthen the lookbacks and the volume window, raise the trend-fit minimum, and expect well under two signals per hour — each one carrying more weight.

As a rule, the most instrument- and timeframe-sensitive settings are the energy minimums, the VPR percentile, and the sequencing rule. Tune those first; leave the rest near default until you have a reason to move them.

10. Tuning recipes

Too many signals / too much noise. Nudge the energy minimums up, raise the VPR percentile, raise Doji Max Body slightly, or step the Signal Sensitivity preset toward Conservative. Turning the energy filter on also thins signals and makes them more climactic.

Too few signals. Lower the energy minimums, lower the VPR percentile, reduce Required Hits (keep it at least 1), or reduce Confirm Seconds Bars. Stepping the preset toward Aggressive does all of this at once.

Direction flips too often. Add a tick or two to the Donchian breakout buffer (Donchian engine) or add a Confirm Seconds Bar (EMA-slope engine), and raise Min Bars After Flip.

Arrows cluster in one spot. Raise Min Bars Between Signals to space them out.

Signals fire inside obvious chop. Confirm the ChopState filter is on, and consider the conditions filter with Regime On Primary to require a clean primary-timeframe trend.

11. Troubleshooting

No arrows at all. Check that you are inside a session window or turn on Ignore Session Filter. Confirm the active gates are not all maximally strict — try the Balanced or Aggressive preset to verify the engine produces signals, then tighten. Remember that several gates on at once is intentionally selective.

The energy filter never seems to trigger. VolDrive's energy term is volume-based. On instruments or feeds without real volume, the energy filter cannot fire — keep Use Energy Filter off there and rely on the other gates.

Arrows seem late. Turning Use Energy Filter on tends to produce later, more climactic signals. Turn it off (energy still feeds tooltips and sequencing) for earlier, continuation-style entries.

Signals disappear or shift after reload. VolDrive calculates on bar close for stable, non-repainting behavior. If you run a multi-series chart, VolDrive may run on the primary series only as a stability safeguard rather than binding to the wrong series; this is expected and consistent.

No sound on a signal. Confirm Sound Alerts is on and that the named WAV files exist in NinjaTrader's sounds folder.

12. Glossary

Continuation signal — an arrow marking a likely continuation of the prevailing move, not a reversal.

Effort — the participation behind a move, read mainly from volume.

Result — whether price actually traveled, read from the candle body and momentum.

Energy — effort and result combined: a push that genuinely moves price.

Energy hit — a bar whose energy clears the active threshold.

Sequencing — the rule that requires enough energy hits within a recent window, so signals reflect repeated push rather than a one-off.

Conditions (market regime) — the overall "weather" check: is the market trending cleanly enough to trade, from efficiency, trend fit and trend strength.

Doji — a bar with a small body and large wicks; an indecisive candle the doji filter can suppress.

ChopState — the detector of sideways congestion (boxes) that VolDrive can use to suppress signals.

Auction phase — the maturity of a move shown in the tooltip: Ignition, Drive or Exhaustion.

Seconds series — a fast secondary data series VolDrive can read for the responsive parts of its work, confirming and drawing on the primary bar.

Flow.42 VolDrive is a context and visualization tool. It does not provide automated trade signals or trading advice. Trade less. Trade better. Trade with clarity.

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